

TEST REPORT

Report No...... : **CHTEW21100065** Report Verification: 

Project No. : **SHT2107068302EW**

Applicant's name : **Chengdu Ebyte Electronic Technology Co., Ltd.**

Address..... : Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China

Test item description..... : **SPI SMD LoRa Module**

Trade Mark..... : EBYTE

Model/Type reference : E22-900M22S

Listed Model(s) : -

Standard : **ETSI EN 300 220-1 V3.1.1:2017-02**
ETSI EN 300 220-2 V3.1.1:2017-02

Date of receipt of test sample..... : Jul.26, 2021

Date of testing..... : Jul.26, 2021- Oct.14, 2021

Date of issue..... : Oct.15, 2021

Result..... : **PASS**

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Testing Laboratory Name : **Shenzhen Huatongwei International Inspection Co., Ltd**

Address..... : 1/F, Bldg 3, Hongfa Hi-tech Industrial Park, Genyu Road,
Tianliao, Gongming, Shenzhen, China

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1. Test standards and Report version

1.1. Test Standards

The tests were performed according to following standards:

[ETSI EN 300 220-1 V3.1.1\(2017-02\)](#)-Short Range Devices (SRD) operating in the frequency range 25 MHz to 1 000 MHz; Part 1: Technical characteristics and methods of measurement

[ETSI EN 300 220-2 V3.1.1\(2017-02\)](#)-Short Range Devices (SRD) operating in the frequency range 25 MHz to 1 000 MHz; Part 2: Harmonised Standard covering the essential requirements of article 3.2 of Directive 2014/53/EU for non specific radio equipment

1.2. Report version information

Revision No.	Date of issue	Description
N/A	2021-10-15	Original

2. Test Description

Transmitter requirement			
Test item	Standards requirement	Result	
		Pass	N/A
Operating frequency	clause 4.2.1	☒	
Unwanted emissions in the spurious domain	clause 4.2.2	☒	
TX Effective radiated power	clause 4.3.1	☒	
TX Maximum e.r.p. spectral density	clause 4.3.2		☒
TX Duty cycle	clause 4.3.3	☒	
TX Occupied bandwidth	clause 4.3.4	☒	
TX Out of band emissions	clause 4.3.5	☒	
TX Transient power	clause 4.3.6	☒	
TX Adjacent channel power	clause 4.3.7		☒
TX Behaviour under low voltage conditions	clause 4.3.8		☒
TX Adaptive power control	clause 4.3.9		☒
TX FHSS	clause 4.3.10		☒
TX Short term behaviour	clause 4.3.11		☒
Receiver requirement			
Test item	Standards requirement	Result	
		Pass	N/A
RX sensitivity	clause 4.4.1		☒
Clear channel assessment threshold	clause 4.5.2		☒
Polite spectrum access timing parameters	clause 4.5.3		☒
RX Blocking	clause 4.4.2	☒	
Adaptive Frequency Agility	clause 4.5.4		☒

Note:

1. N/A is not application
2. For applicable test item please see ETSI EN 300 220-2 Annex A EN Requirements Table
3. EUT is belong to ETSI EN 300 220-2 Annex C XBand

3. Summary

3.1. Client Information

Applicant:	Chengdu Ebyte Electronic Technology Co., Ltd.
Address:	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China
Manufacturer:	Chengdu Ebyte Electronic Technology Co., Ltd.
Address:	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China
Factory:	Chengdu Ebyte Electronic Technology Co., Ltd.
Address:	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China

3.2. Product Description

Name of EUT:	SPI SMD LoRa Module		
Trade Mark:	EBYTE		
Model/Type reference	E22-900M22S		
Listed Model(s):	-		
Power supply:	DC 5V		
Transmitter unit:			
Operation Frequency:	863MHz~870MHz		
Modulation type:	LORA		
Modulation connector:	☒ Without external ☐ External		
Occupied bandwidth ^{#1} :	125KHz		
Product type:	☒ Wideband Tranceiver ☐ Narrowband Tranceiver		
Antenna type:	Sucker antenna		
Antenna gain ^{#2} :	3dBi		
Hardware Version:	V1.0		
Software Version:	V1.0		
Receiver unit:			
Receiver category ^{#3} :	☒ 2	☐ 1.5	☐ 1

Note:

#1: Declared by the manufacturer

#2: The antenna gain is provided by the applicant, and the applicant should be responsible for its authenticity, HTW lab has not verified the authenticity of its information.

#3: Declared by the manufacturer

3.3. Testing Laboratory Information

Laboratory Name	Shenzhen Huatongwei International Inspection Co., Ltd.
Laboratory Location	1/F, Bldg 3, Hongfa Hi-tech Industrial Park, Genyu Road, Tianliao, Gongming, Shenzhen, China
Connect information:	Tel: 86-755-26715499 E-mail: cs@szhtw.com.cn http://www.szhtw.com.cn

3.4. Modifications

No modifications were implemented to meet testing criteria.

4. Test Configuration

4.1. Test frequency list

Test Channel	Frequency (MHz)
CH-L	863.1
CH-M	868.3
CH-H	869.9

4.2. Test mode

Test mode	Transmitting	Receiving
TX	√	
RX		√

√: is operation mode and modulation type .

Test mode	Test item
TX	TX Occupied bandwidth, TX Effective radiated power, TX Transient power, Tx Out Of Band Emissions, Unwanted emissions in the spurious domain, TX Duty cycle
RX	Blocking, Unwanted emissions in the spurious domain

4.3. EUT configuration

The following peripheral devices and interface cables were connected during the measurement:

- - supplied by the manufacturer
- - supplied by the lab

1	Laptop	Manufacturer :	DELL
		Model No. :	Inspiron 13-5378
		Manufacturer :	/
		Model No. :	/

4.4. Test conditions

Normal Conditon	Temperature	15 °C to 35 °C	
	Relative humidity	20 % to 75 %.	
	Voltage	☒ Mains voltage	Nominal mains voltage
		☐ USB 5V power supply	
		☐ Lead-acid battery	1.1 * the nominal voltage of the battery
☐ Other		the normal test voltage shall be that declared by the equipment provider	
Extreme Conditon	Temperature	☐ The temperature range as declared by the provider	
		☒ -40 °C to +70°C for Temperature category I (General)	
		☐ -10 °C to +55 °C for Temperature category II (Portable)	
		☐ +5 °C to +35 °C for Temperature category III (Equipment for normal indoor use)	
	Voltage	☒ Mains voltage	±10 %* the nominal mains voltage
		☐ Lead-acid battery	1,3 and 0,9 multipliyed by the nominal voltage of the battery
		☐ Leclanché or the lithium battery	Lower extreme voltage: 0.85*the nominal voltage upper extreme voltage: declared by the equipment provider
		☐ Nickel-cadmium battery	Lower extreme voltage: 0.9*the nominal voltage upper extreme voltage: declared by the equipment provider
		☐ Other	the Extreme test voltage shall be that declared by the equipment provider
Normal Conditon		V _N =Nominal Voltage	DC 5.0V
		T _N =Normal Temperature	25 °C
Extreme Conditon		V _L =Lower Voltage	DC 4.5V
		T _L =Lower Temperature	-40°C
		V _H =Higher Voltage	DC 5.5V
		T _H =Higher Temperature	85C

4.5. Statement of the measurement uncertainty

Test Items	Measurement Uncertainty	Notes
Frequency range	70Hz for <1GHz 130Hz for >1GHz	(1)
Occupied Bandwidth	70Hz for <1GHz 130Hz for >1GHz	(1)
Transmitter power conducted	0.77 dB	(1)
Conducted spurious emissions 9kHz~40GHz	0.77 dB	(1)
Radiated spurious emissions	4.36dB for <1GHz 5.10dB for >1GHz	(1)
Adjacent Channel Selectivity	1.25 dB	(1)
Blocking	1.91 dB	(1)

(1) This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=1.96.

4.6. Equipments Used during the Test

● Radiated emission-6th test site							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Semi-Anechoic Chamber	Albatross projects	HTWE0127	SAC-3m-02	C11121	2018/09/30	2022/09/29
●	EMI Test Receiver	R&S	HTWE0099	ESCI	100900	2021/9/14	2022/9/13
●	Loop Antenna	R&S	HTWE0170	HFH2-Z2	100020	2021/04/06	2022/04/05
●	Ultra-Broadband Antenna	SCHWARZBECK	HTWE0123	VULB9163	538	2021/04/06	2022/04/05
●	Pre-Amplifier	SCHWARZBECK	HTWE0295	BBV 9742	N/A	2020/11/13	2021/11/12
●	RF Connection Cable	HUBER+SUHNER	HTWE0062-01	N/A	N/A	2021/02/26	2022/02/25
●	RF Connection Cable	HUBER+SUHNER	HTWE0062-02	SUCOFLEX104	501184/4	2021/02/26	2022/02/25
●	Test Software	R&S	N/A	ES-K1	N/A	N/A	N/A

● Radiated emission-7th test site							
Used	Test Equipment	Manufacturer	Equipment No.	Model No.	Serial No.	Last Cal. Date (YY-MM-DD)	Next Cal. Date (YY-MM-DD)
●	Semi-Anechoic Chamber	Albatross projects	HTWE0122	SAC-3m-01	N/A	2018/09/27	2022/09/26
●	Spectrum Analyzer	R&S	HTWE0098	FSP40	100597	2021/9/13	2022/9/12
●	Horn Antenna	SCHWARZBECK	HTWE0126	9120D	1011	2020/04/01	2023/03/31
●	Broadband Horn Antenna	SCHWARZBECK	HTWE0103	BBHA9170	BBHA9170472	2020/4/27	2023/4/27
●	Pre-amplifier	CD	HTWE0071	PAP-0102	12004	2020/11/13	2021/11/12
●	Broadband Pre-amplifier	SCHWARZBECK	HTWE0201	BBV 9718	9718-248	2021/03/05	2022/03/04
●	RF Connection Cable	HUBER+SUHNER	HTWE0120-01	6m 18GHz S Serisa	N/A	2021/02/26	2022/02/25
●	RF Connection Cable	HUBER+SUHNER	HTWE0120-02	6m 3GHz RG Serisa	N/A	2021/02/26	2022/02/25
●	RF Connection Cable	HUBER+SUHNER	HTWE0119-05	6m 3GHz RG Serisa	N/A	2021/02/26	2022/02/25
●	RF Connection Cable	HUBER+SUHNER	HTWE0120-04	6m 3GHz RG Serisa	N/A	2021/02/26	2022/02/25
●	RF Connection Cable	HUBER+SUHNER	HTWE0121-01	6m 18GHz S Serisa	N/A	2021/02/26	2022/02/25
●	Test Software	Audix	N/A	E3	N/A	N/A	N/A

● Conducted method						
Item	Test Equipment	Manufacturer	Model No.	Serial No.	Last Cal. (mm-dd-yy)	Next Cal. (mm-dd-yy)
●	Spectrum analyzer	R&S	FSV40	100048	2020/10/19	2021/10/18
●	Signal generator	R&S	SMB100A	1406.600K0 3-177956-Uh	2020/10/19	2021/10/18
●	Power Divider	Ocean Microwave	OPD1060-4-SMA	N/A	N/A	N/A

5. Test conditions and Results

5.1. Transmitter requirement

5.1.1. Operating frequency

The nominal Operating Frequency is the centre of a channel of width OCW. Declared by the manufacturer.

LIMIT

ETSI EN 300 220-1 Sub-clause 5.1.2

Value	Notes
Operational Frequency band or bands	Declared by the manufacturer
Nominal Operating Frequency or Frequencies	Declared by the manufacturer
Operating Channel width(s) - OCW	Declared by the manufacturer

Operating channel(s) shall be be entirely within operational frequency bands allowed by EN 300220-2- annexes B, C or any NRI.

TEST CONFIGURATION

N/A

TEST PROCEDURE

- The test conditions.
☒ normal condition ☐ Extreme conditions
- Declared by the manufacturer.

TEST RESULTS

☒ Passed ☐ Not Applicable

Test item	Value	
Operational Frequency band or bands	X band	863 MHz to 870 MHz
Nominal Operating Frequency or Frequencies	863.1 MHz to 869.9 MHz,	
Operating Channel width(s) - OCW	125 kHz	

5.1.2. TX Effective radiated power

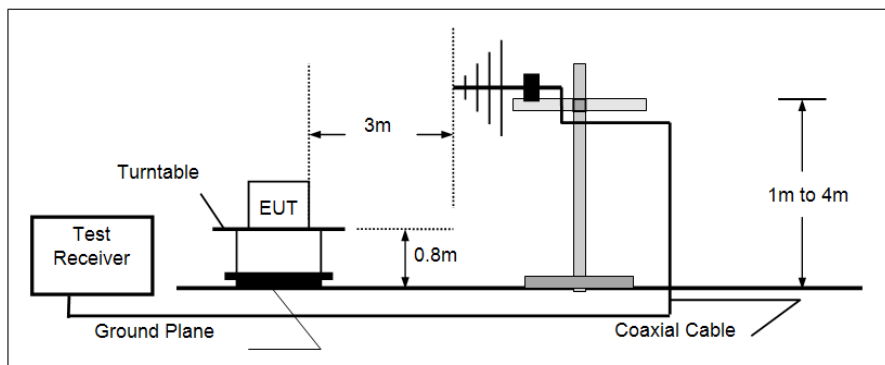
The effective radiated power (e.r.p) is the power radiated in the direction of the maximum radiated power under specified conditions of measurements for any condition of modulation. For equipment with a permanent or temporary antenna connection it may be taken as the power delivered from that connector taking into account the antenna gain.

LIMIT

ETSI EN 300 220-2 The effective radiated power shall not be greater than the value allowed in annexes B or C for the chosen operational frequency band(s).

Operational Frequency Band		Effective Radiated Power	duty cycle
X	863 MHz to 870 MHz	25 mW	<0.1%

TEST CONFIGURATION



TEST PROCEDURE

- The test conditions.
☒ normal condition ☐ Extreme conditions
- Refer to ETSI EN 300 220-1 Sub-clause 5.2.2.1.2 for the measurement method.

TEST RESULTS

☒ Passed ☐ Not Applicable

Please refer to appendix A on the appendix report

5.1.3. TX Maximum e.r.p. spectral density

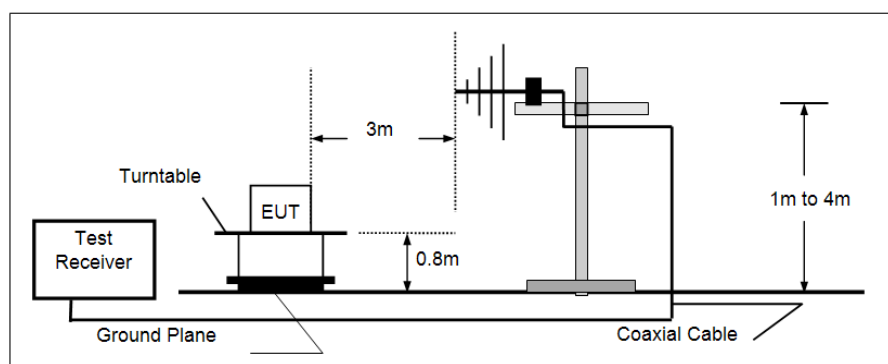
Maximum e.r.p. power spectral density applies to transmitters using DSSS or wideband techniques other than FHSS modulation, in annex C band

LIMIT

ETSI EN 300 220-2 The effective radiated power shall not be greater than the value allowed in annexes B or C for the chosen operational frequency band(s).

Operational Frequency Band		Effective Radiated Power	TX Maximum e.r.p. spectral density
X	863 MHz to 870 MHz	25 mW	N/A

TEST CONFIGURATION



TEST PROCEDURE

- The test conditions.
☒ normal condition ☐ Extreme conditions
- Refer to ETSI EN 300 220-1 Sub-clause 5.3.2.1.2 for the measurement method.

TEST RESULTS

☐ Passed ☒ Not Applicable

5.1.4. TX Occupied Bandwidth

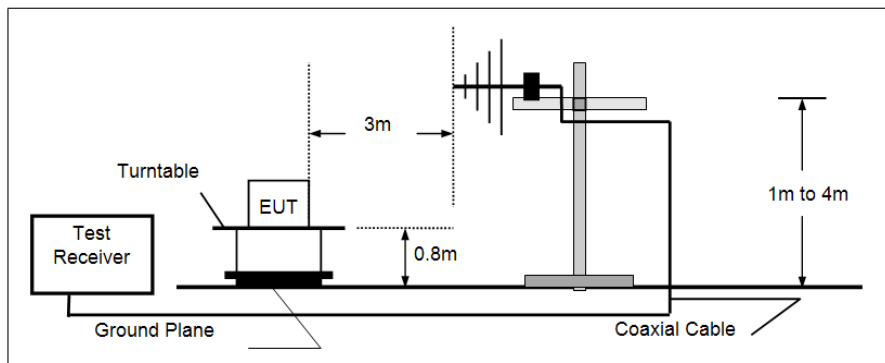
The occupied bandwidth (OBW) is the Frequency Range in which 99 % of the total mean power of a given emission falls.

LIMIT

ETSI EN 300 220-1 Sub-clause 5.6.2

The Operating Channel shall be declared and shall reside entirely within the Operational Frequency Band. The Maximum Occupied Bandwidth at 99 % shall reside entirely within the Operating Channel defined by F_{low} and F_{high} .

TEST CONFIGURATION



TEST PROCEDURE

3. Refer to ETSI EN 300 220-1 Sub-clause 5.6.3.3 for the measurement method.

TEST RESULTS

☒ Passed ☐ Not Applicable

Please refer to appendix B on the appendix report

5.1.5. Tx Out Of Band Emissions

Two OOB domains are defined, one for OC (ETSI EN 300 220-1 see Figure 5) and one for Operational Frequency band (ETSI EN 300 220-1 see Figure 6). The spectrum masks for these two OOB domains may overlap.

LIMIT

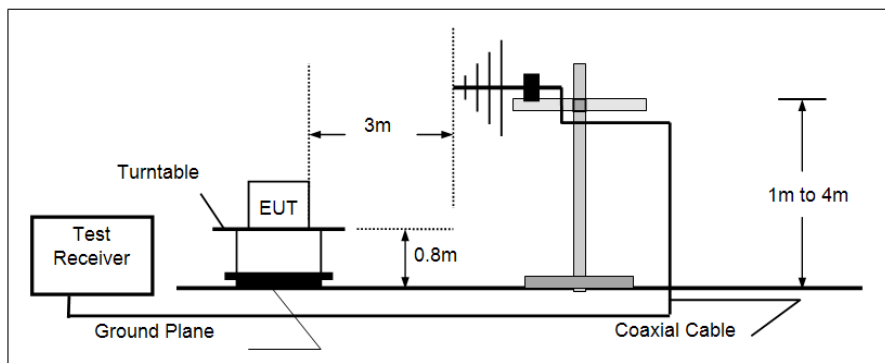
ETSI EN 300 220-1 Sub-clause 5.8.2

Domain	Frequency Range	RBW _{REF}	Max power limit
OOB limits applicable to Operational Frequency Band (See Figure 6)	$f \leq f_{\text{low_OFB}} - 400 \text{ kHz}$	10 kHz	-36 dBm
	$F_{\text{low_OFB}} - 400 \text{ kHz} \leq f \leq f_{\text{low_OFB}} - 200 \text{ kHz}$	1 kHz	-36 dBm
	$f_{\text{low}} - 200 \text{ kHz} \leq f < f_{\text{low_OFB}}$	1 kHz	See Figure 6
	$f = f_{\text{low_OFB}}$	1 kHz	0 dBm
	$f = f_{\text{high_OFB}}$	1 kHz	0 dBm
	$F_{\text{high_OFB}} < f \leq f_{\text{high_OFB}} + 200 \text{ kHz}$	1 kHz	See Figure 6
	$F_{\text{high_OFB}} + 200 \text{ kHz} \leq f \leq f_{\text{high_OFB}} + 400 \text{ kHz}$	1 kHz	-36 dBm
	$F_{\text{high_OFB}} + 400 \text{ kHz} \leq f$	10 kHz	-36 dBm
OOB limits applicable to Operating Channel (See Figure 5)	$f = f_c - 2.5 \times \text{OCW}$	1 kHz	-36 dBm
	$f_c - 2.5 \times \text{OCW} \leq f \leq f_c - 0.5 \times \text{OCW}$	1 kHz	See Figure 5
	$f = f_c - 0.5 \times \text{OCW}$	1 kHz	0 dBm
	$f = f_c + 0.5 \times \text{OCW}$	1 kHz	0 dBm
	$f_c + 0.5 \times \text{OCW} \leq f \leq f_c + 2.5 \times \text{OCW}$	1 kHz	See Figure 5
	$f = f_c + 2.5 \times \text{OCW}$	1 kHz	-36 dBm

NOTE: f is the measurement frequency.
 f_c is the Operating Frequency.
 $F_{\text{low_OFB}}$ is the lower edge of the Operational Frequency Band.
 $F_{\text{high_OFB}}$ is the upper edge of the Operational Frequency Band.
OCW is the operating channel bandwidth.

Note: See Figure 5/6 is reference for ETSI EN 300 220-1

TEST CONFIGURATION



TEST PROCEDURE

- The test conditions.
 - ☒ normal condition
 - ☒ Extreme conditions
- Refer to ETSI EN 300 220-1 Sub-clause 5.8.3.3 for the measurement method.

TEST RESULTS

☒ Passed ☐ Not Applicable

TX Out of Band Emissions applies to all transmitters with OCW > 25 kHz.

Please refer to appendix C on the appendix report

5.1.6. TX Transient power

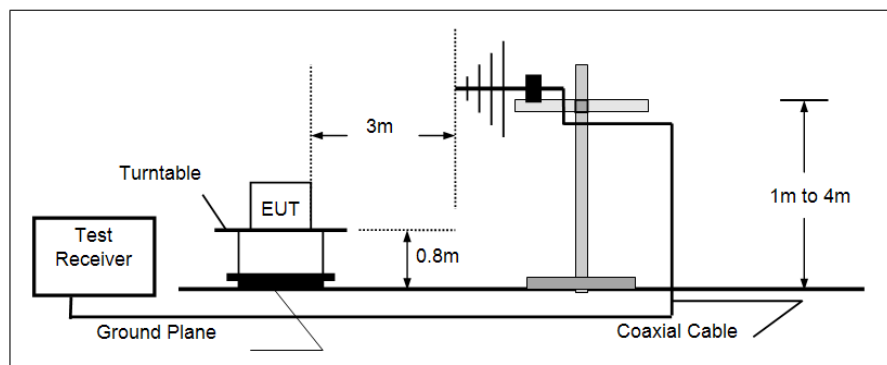
Transmitter transient power is power falling into frequencies other than the operating channel as a result of the transmitter being switched on and off.

LIMIT

ETSI EN 300 220-1 Sub-clause 5.10.2

Absolute offset from centre frequency	RBW _{REF}	Peak power limit applicable at measurement points
≤ 400 kHz	1 kHz	0 dBm
> 400 kHz	1 kHz	-27 dBm

TEST CONFIGURATION



TEST PROCEDURE

- The test conditions.
☒ normal condition ☐ Extreme conditions
- Refer to ETSI EN 300 220-1 Sub-clause 5.10.3.2 for the measurement method.

TEST RESULTS

☒ Passed ☐ Not Applicable

Please refer to appendix D on the appendix report

5.1.7. Unwanted emissions in the spurious domain(Conducted)

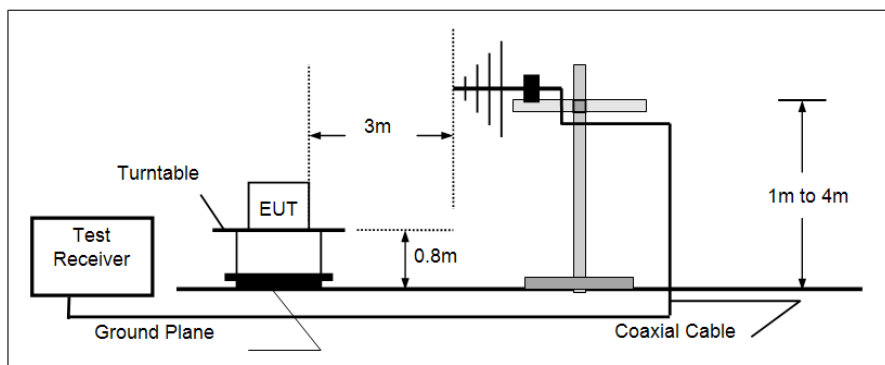
Spurious emissions are unwanted emissions in the spurious domain are emissions at frequencies other than those of the wanted carrier frequency and its sidebands associated with normal test modulation

LIMIT

ETSI EN 300 220-1 Sub-clause 5.9.2

Frequency	47 MHz to 74 MHz 87,5 MHz to 118 MHz 174 MHz to 230 MHz 470 MHz to 790 MHz	Other frequencies below 1 000 MHz	Frequencies above 1 000 MHz
State			
TX mode	-54 dBm	-36 dBm	-30 dBm
RX and all other modes	-57 dBm	-57 dBm	-47 dBm

TEST CONFIGURATION



TEST PROCEDURE

- The test conditions.
☒ normal condition ☐ Extreme conditions
- Refer to ETSI EN 300 220-1 Sub-clause 5.9.3.3.1 for the measurement method.

TEST RESULTS

☒ Passed ☐ Not Applicable

Please refer to appendix E and F on the appendix report

5.1.8. Unwanted emissions in the spurious domain(Radiated)

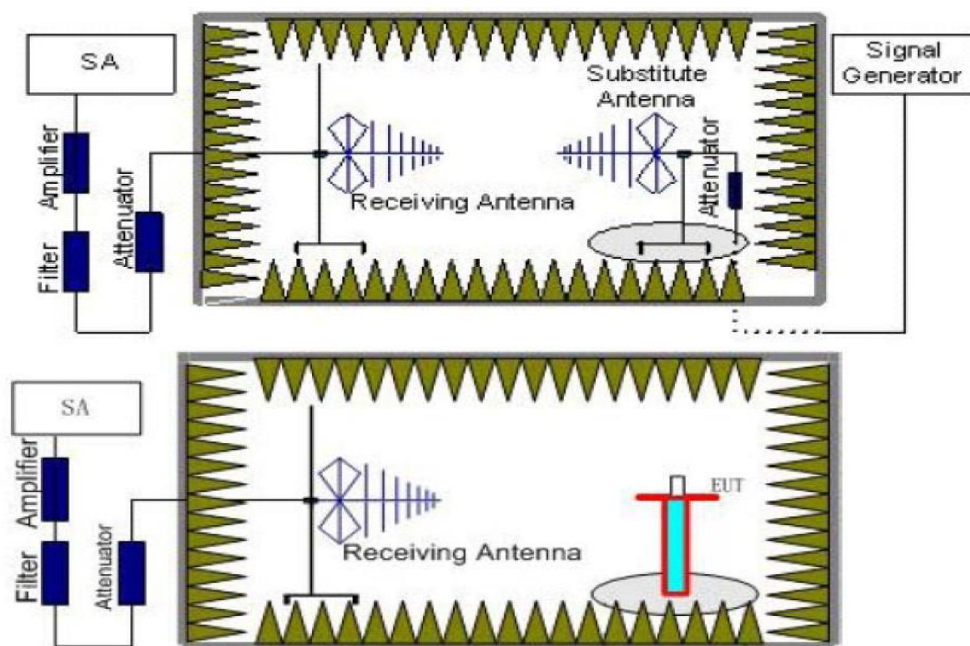
Spurious emissions are unwanted emissions in the spurious domain are emissions at frequencies other than those of the wanted carrier frequency and its sidebands associated with normal test modulation

LIMIT

ETSI EN 300 220-1 Sub-clause 5.9.2

Frequency	47 MHz to 74 MHz 87,5 MHz to 118 MHz 174 MHz to 230 MHz 470 MHz to 790 MHz	Other frequencies below 1 000 MHz	Frequencies above 1 000 MHz
State			
TX mode	-54 dBm	-36 dBm	-30 dBm
RX and all other modes	-57 dBm	-57 dBm	-47 dBm

TEST CONFIGURATION



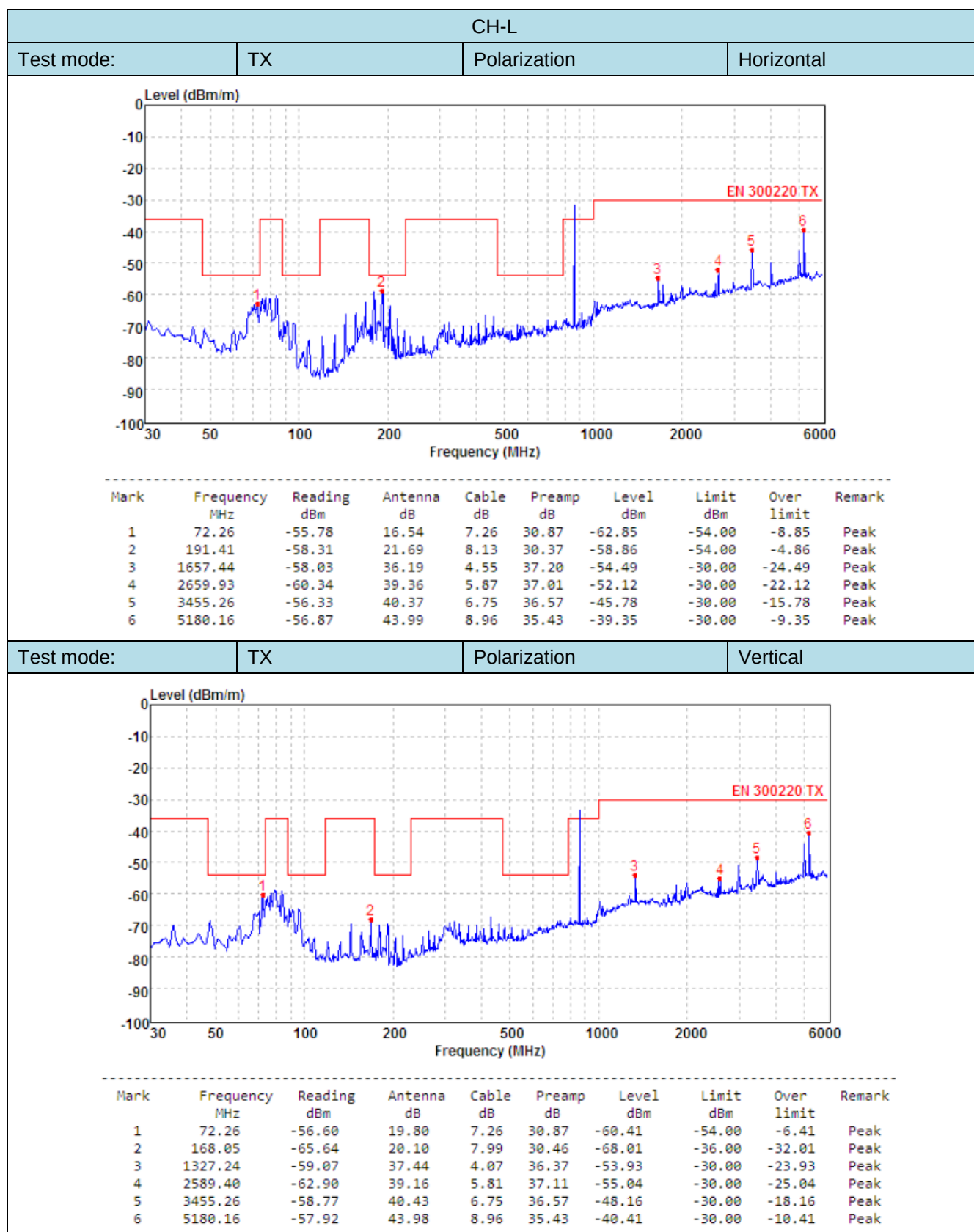
TEST PROCEDURE

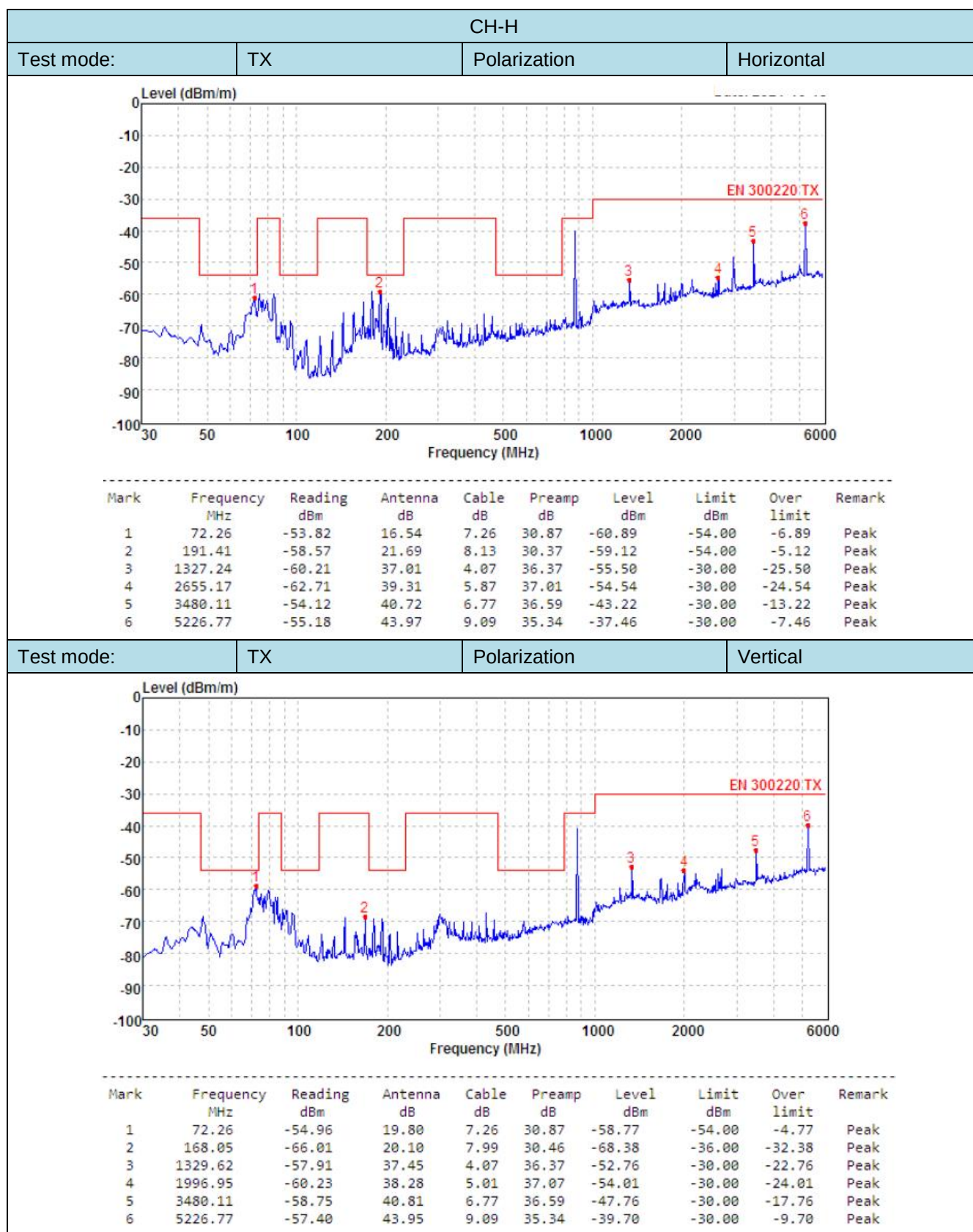
- The test conditions.
☒ normal condition ☐ Extreme conditions
- Refer to ETSI EN 300 220-1 Sub-clause 5.9.3.3.2 for the measurement method.

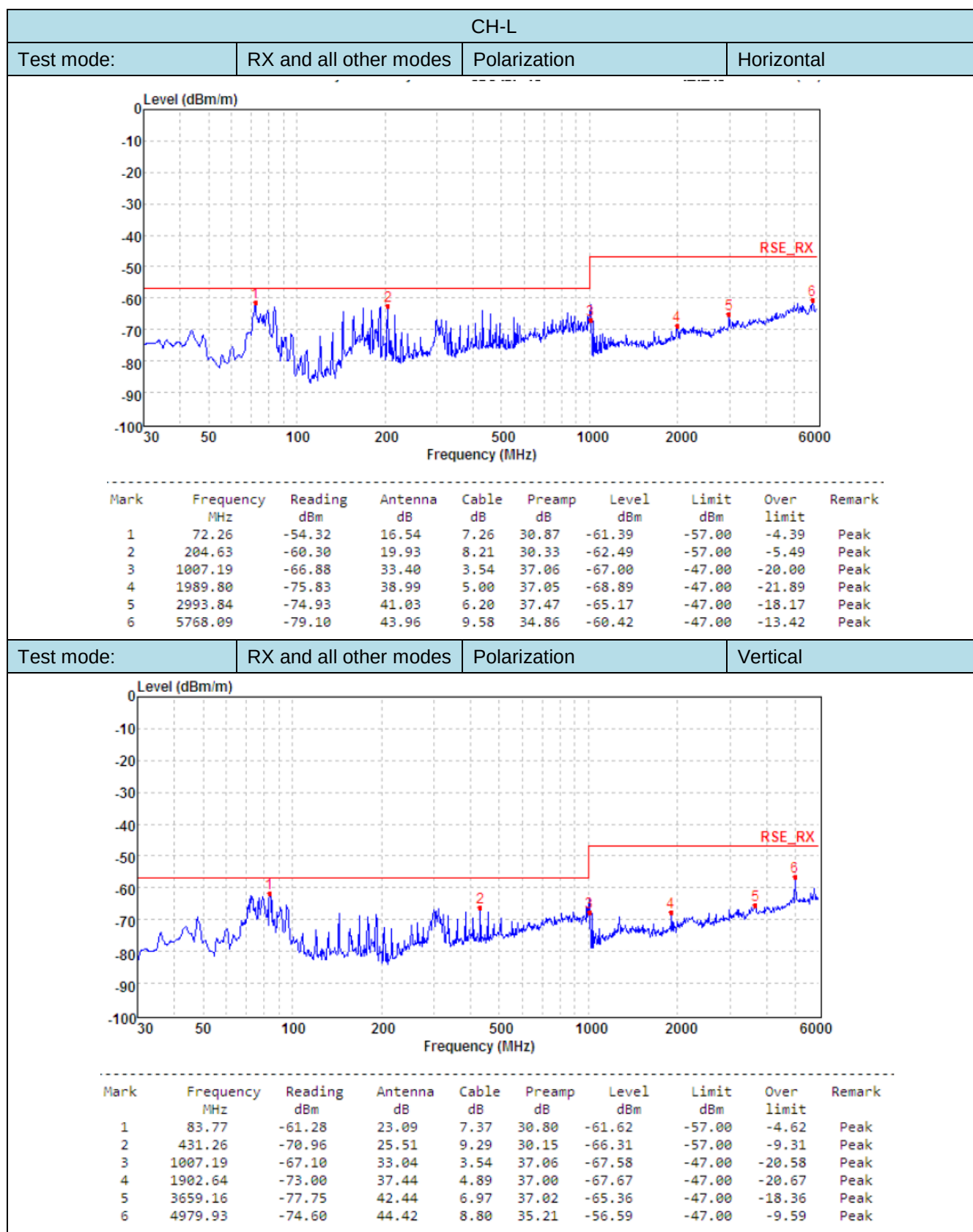
TEST RESULTS

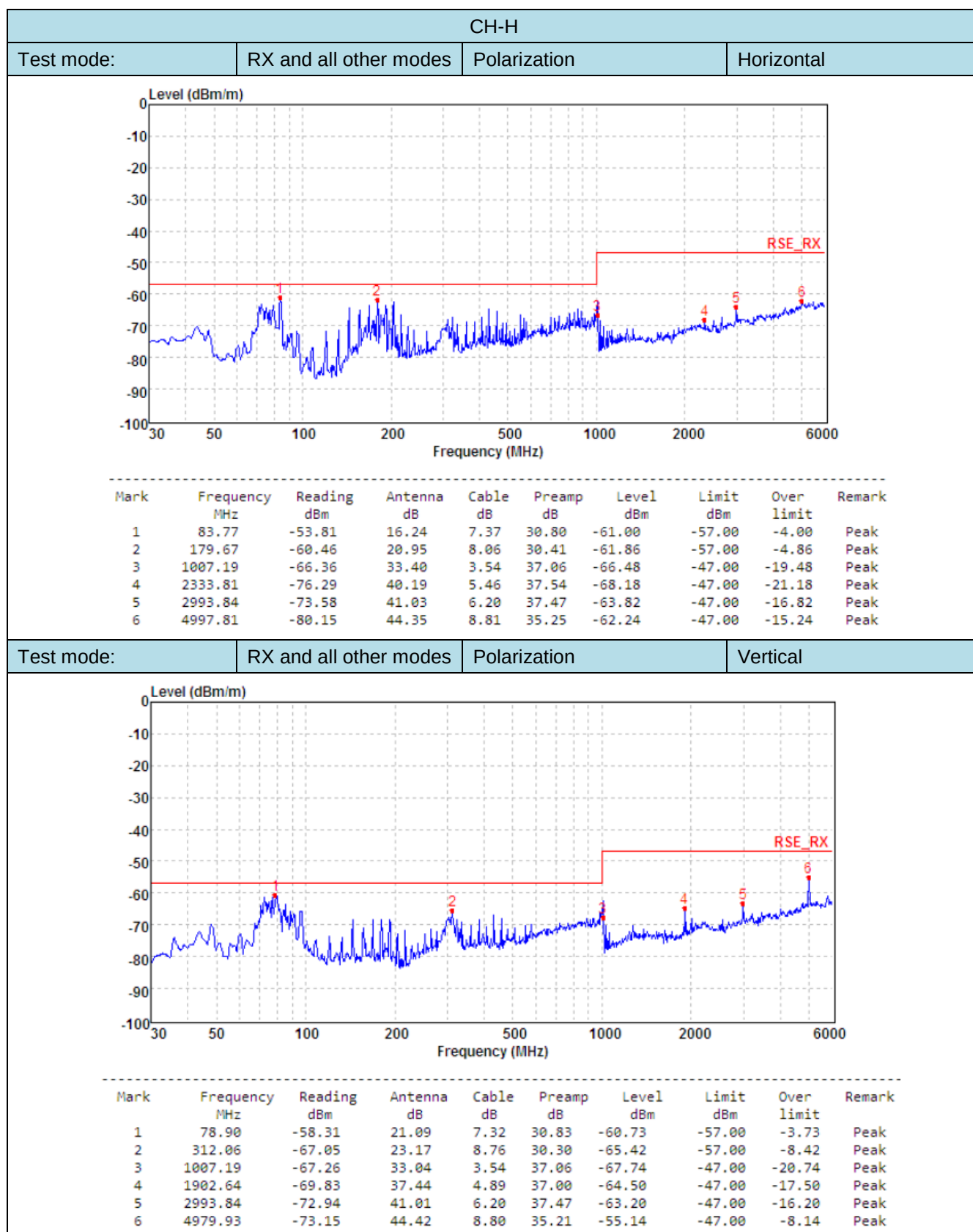
☒ Passed ☐ Not Applicable

Please refer to the below test data:









5.1.9. TX Behaviour under Low Voltage Conditions

The TX behaviour under low voltage condition is the ability of the equipment to maintain its operating frequency and not produce emissions which exceed any relevant limit when the battery voltage falls below the lower extreme voltage level.

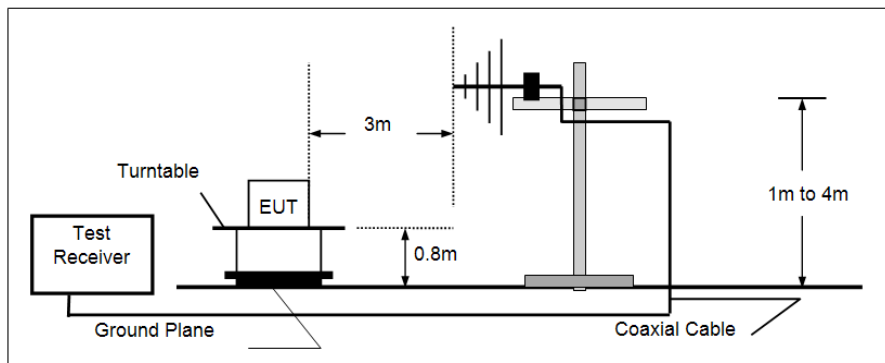
LIMIT

ETSI EN 300 220-1 Sub-clause 5.12.2

The equipment shall either:

- a) remain in the Operating Channel OC without exceeding any applicable limits (e.g. Duty Cycle); or
- b) reduce its effective radiated power below the Spurious Emission limits without exceeding any applicable limits (e.g. Duty Cycle); or
- c) shut down, (ceasing function); as the voltage falls below the manufacturers declared operating voltage.

TEST CONFIGURATION



TEST PROCEDURE

1. The test conditions.
 - ☒ normal condition
 - ☐ Extreme conditions
2. Refer to ETSI EN 300 220-1 Sub-clause 5.12.3.2 for the measurement method.

TEST RESULTS

☐ Passed ☒ Not Applicable

Note: The EUT is powered by DC power (no battery)

5.1.10. TX Duty Cycle

Duty cycle is the ratio expressed as a percentage, of the cumulative duration of transmissions T_{on_cum} within

$$DC = \left(\frac{T_{on_cum}}{T_{obs}} \right) F_{obs}$$

an observation interval T_{obs} . F_{obs} on an observation bandwidth F_{obs} .

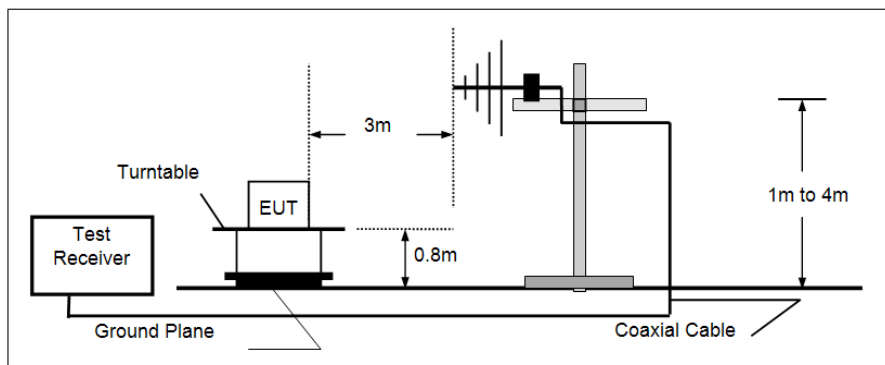
Unless otherwise specified, T_{obs} is 1 hour and the observation bandwidth F_{obs} is the operational frequency band. Each transmission consists of an RF emission, or sequence of RF emissions separated by intervals $< T_{Dis}$.

LIMIT

ETSI EN 300 220-2 The effective radiated power shall not be greater than the value allowed in annexes B or C for the chosen operational frequency band(s).

Operational Frequency Band		duty cycle
X	863 MHz to 870 MHz	<0.1 %

TEST CONFIGURATION



TEST PROCEDURE

Refer to ETSI EN 300 220-1 Sub-clause 7.10.2 for the declaration.

TEST RESULTS

☒ Passed ☐ Not Applicable

Please refer to the below test data:

The duty cycle was not exceeded 0.1% in a period of 1 hour, which was declared by the manufacturer. EUT was transmitting approximately 0.1s(maximum) at a time, so it will be restricted transmitting less than 360 times in a period of 1 hour.

5.2. Receiver requirements

5.2.1. Blocking

Blocking is a measure of the capability of the receiver to receive a wanted modulated signal without exceeding a given degradation due to the presence of an unwanted input signal at any frequencies other than those of the spurious responses or the adjacent channels or bands.

LIMIT

ETSI EN 300 220-1 Sub-clause 5.18.2

2.Reference limits for receiver category 2

Requirement	Limits
	Receiver category 2
Blocking at ± 2 MHz from OC edge f_{high} and f_{low}	≥ -69 dBm
Blocking at ± 10 MHz from OC edge f_{high} and f_{low}	≥ -44 dBm
Blocking at ± 5 % of Centre Frequency or 15 MHz, whichever is the greater	≥ -44 dBm

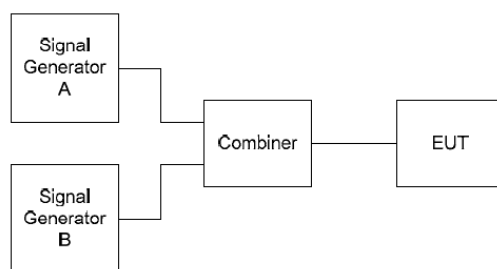
3. Reference limits for receiver category 1.5

Requirement	Limits
	Receiver category 1.5
Blocking at ± 2 MHz from OC edge f_{high} and f_{low}	≥ -43 dBm
Blocking at ± 10 MHz from OC edge f_{high} and f_{low}	≥ -33 dBm
Blocking at ± 5 % of Centre Frequency or 15 MHz, whichever is the greater	≥ -33 dBm

4. Reference limits for receiver category 1

Requirement	Limits
	Receiver category 1
Blocking at ± 2 MHz from Centre Frequency	≥ -20 dBm
Blocking at ± 10 MHz from Centre Frequency	≥ -20 dBm
Blocking at ± 5 % of Centre Frequency or 15 MHz, whichever is the greater	≥ -20 dBm

TEST CONFIGURATION



TEST PROCEDURE

- The test conditions.
☒ normal condition ☐ Extreme conditions
- Please refer to ETSI EN 300 220-1 Sub-clause 5.6.3 for the measurement method.

TEST RESULTS

☒ Passed ☐ Not Applicable

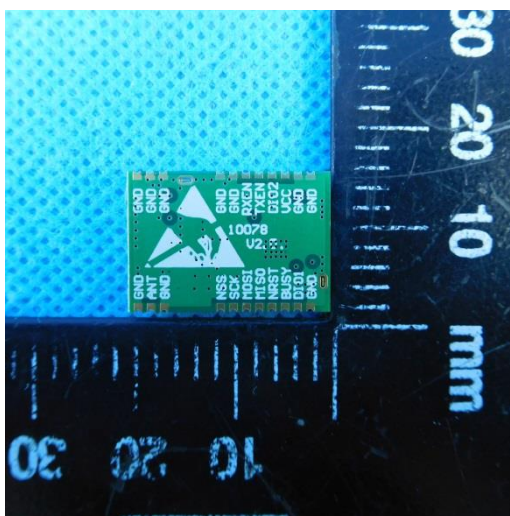
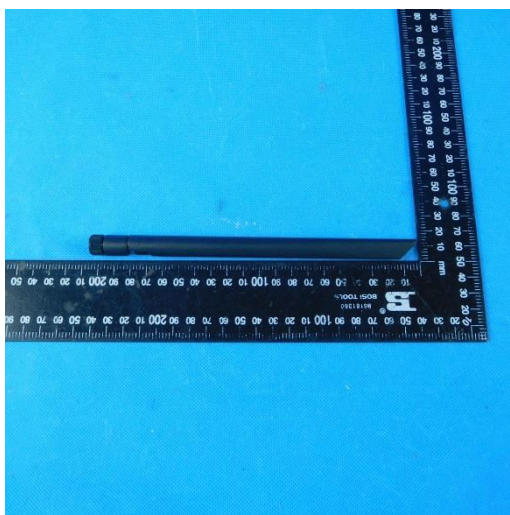
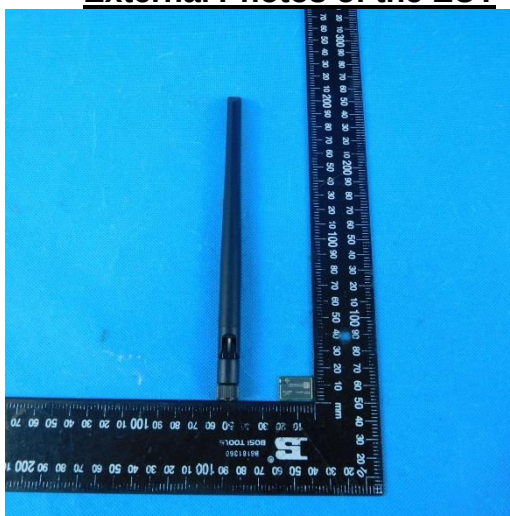
Please refer to appendix G on the appendix report

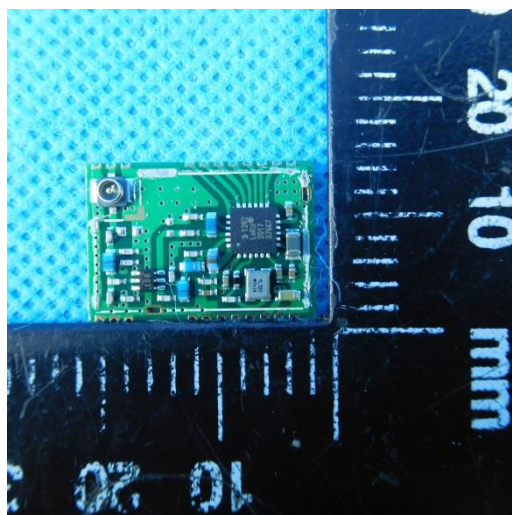
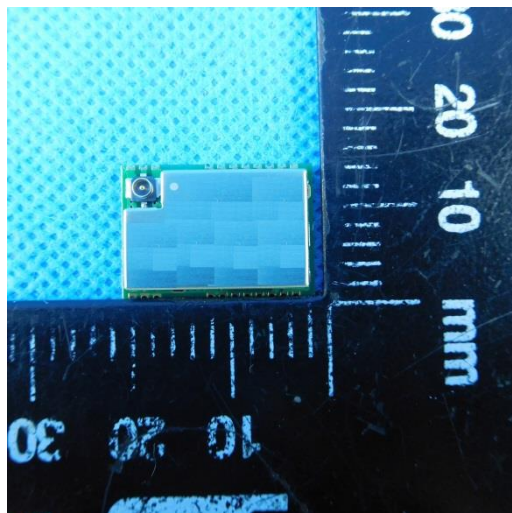
6. Test Setup Photos of the EUT



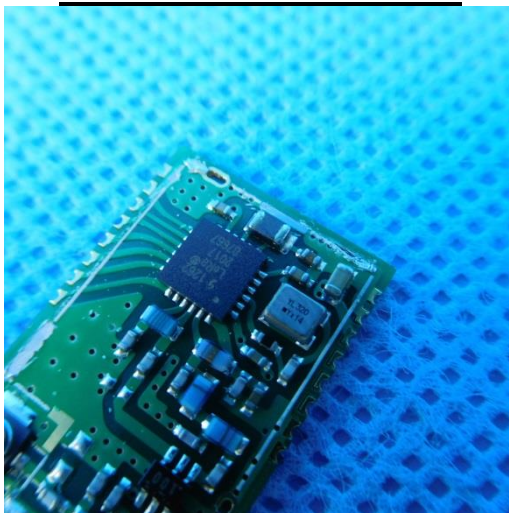
7. External and Internal Photos of the EUT

External Photos of the EUT





Internal Photos of the EUT



8. Appendix

APPENDIX REPORT

Project No.	SHT2107068302EW	Radio Specification	LORA
Test sample No.	YPHT21070683002	Model No.	E22-900M22S
Start test date	2021-08-04	Finish date	2021-08-04
Test Engineer	Qizhi Zhang	Auditor	Xiaodong Zheo

Appendix clause	Test item	Result
A	Effective Radiated Power	PASS
B	Occupied Bandwidth	PASS
C	Out Of Band Emissions	PASS
D	Transient Power	PASS
E	Transmitter Unwanted Emissions in the Spurious Domain	PASS
F	Receiver Spurious Emissions	PASS
G	Blocking	PASS

Appendix A: Effective Radiated Power

Test conditions		Test Channel (MHz)	Conducted Power (dBm)	Antenna gain (dBi)	ERP (dBm)	Limit (dBm)	Result
Temperature	Voltage						
TN	VN	CH-L	8.24	3	9.09	14.00	Pass
		CH-M	7.93	3	8.78		
		CH-H	7.81	3	8.66		
TL	VL	CH-L	9.17	3	10.02	14.00	Pass
		CH-M	8.48	3	9.33		
		CH-H	7.96	3	8.81		
	VH	CH-L	8.38	3	9.23	14.00	Pass
		CH-M	8.36	3	9.21		
		CH-H	7.45	3	8.30		
TH	VL	CH-L	7.72	3	8.57	14.00	Pass
		CH-M	7.47	3	8.32		
		CH-H	6.82	3	7.67		
	VH	CH-L	8.15	3	9.00	14.00	Pass
		CH-M	7.99	3	8.84		
		CH-H	7.53	3	8.38		

Note:

E.R.P=conducted power + Antenna gain (dBd)

Antenna gain (dBd)=Antenna gain (dBi)-2.15

Appendix B: Occupied Bandwidth

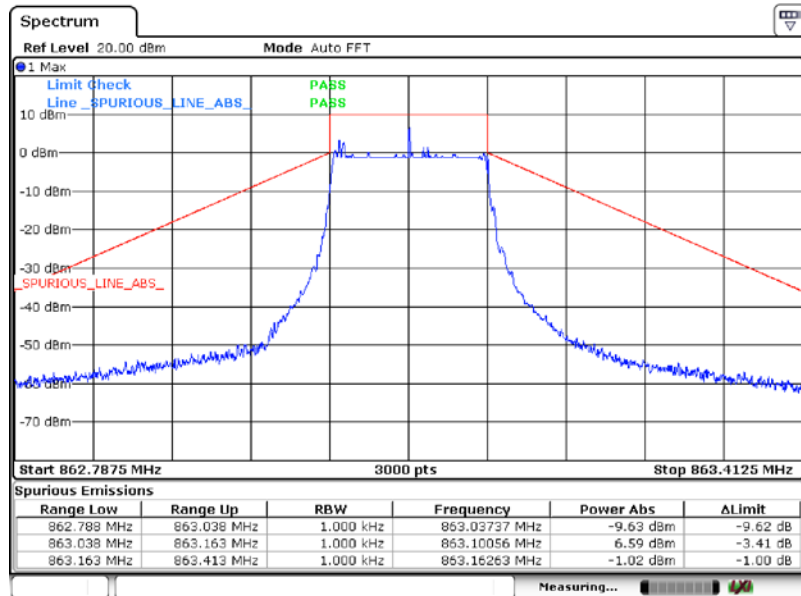
Test conditions		Test Channel (MHz)	Occupied bandwidth (kHz)	f_L (MHz)	f_H (MHz)	Limit	Result
Temper	Voltage						
T_N	V_N	CH-L	125.90	863.0378	863.1637	915.2-920.8	Pass
		CH-H	125.90	869.8378	869.9637		
T_L	V_L	CH-L	125.93	863.0378	863.1637	915.2-920.8	Pass
		CH-H	125.91	869.8378	869.9637		
	V_H	CH-L	125.89	863.0378	863.1637	915.2-920.8	Pass
		CH-H	125.91	869.8378	869.9637		
T_H	V_L	CH-L	125.95	863.0378	863.1637	915.2-920.8	Pass
		CH-H	125.92	869.8378	869.9637		
	V_H	CH-L	125.90	863.0378	863.1637	915.2-920.8	Pass
		CH-H	125.93	869.8378	869.9637		

Appendix C: Out Of Band Emissions

Test item	Test Channel	Result
Out Of Band Domain for Operating Channel	CH-L	PASS
	CH-M	PASS
	CH-H	PASS
Out Of Band Domain for Operational Frequency Band	CH-L	PASS
	CH-H	PASS

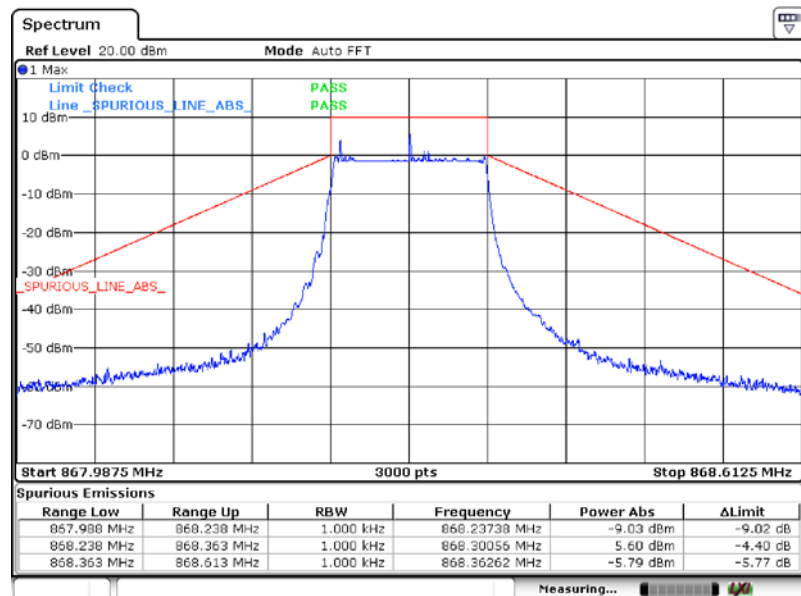
Out Of Band Domain for Operating Channel

CH-L



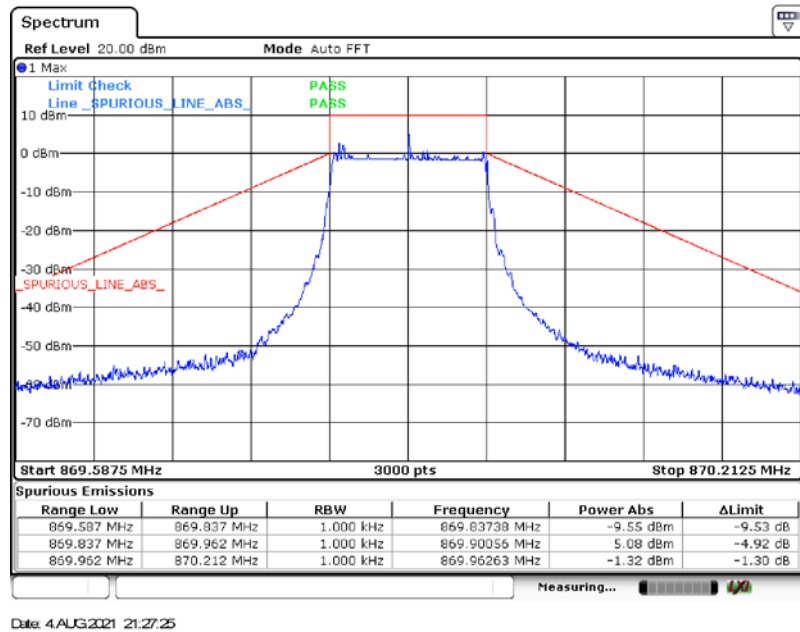
Date: 4 AUG 2021 21:21:49

CH-M



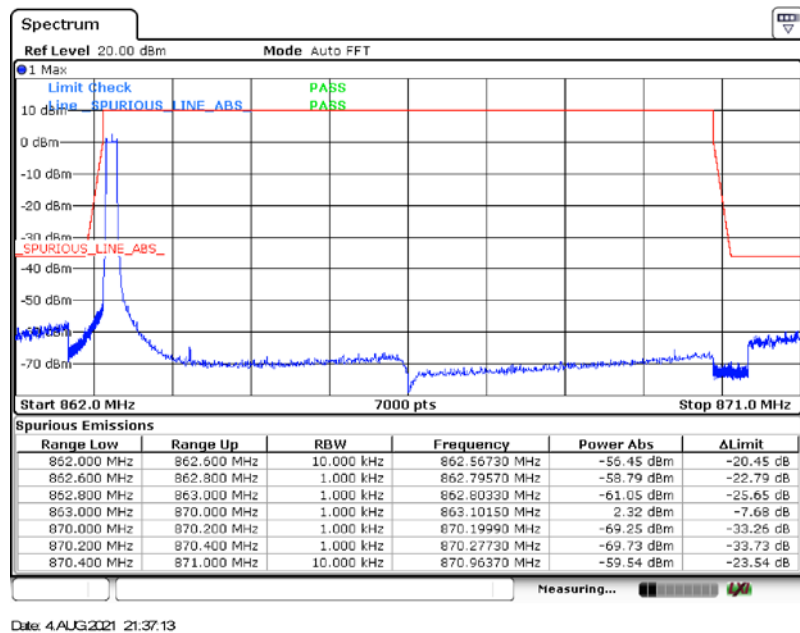
Date: 4 AUG 2021 21:24:52

CH-H

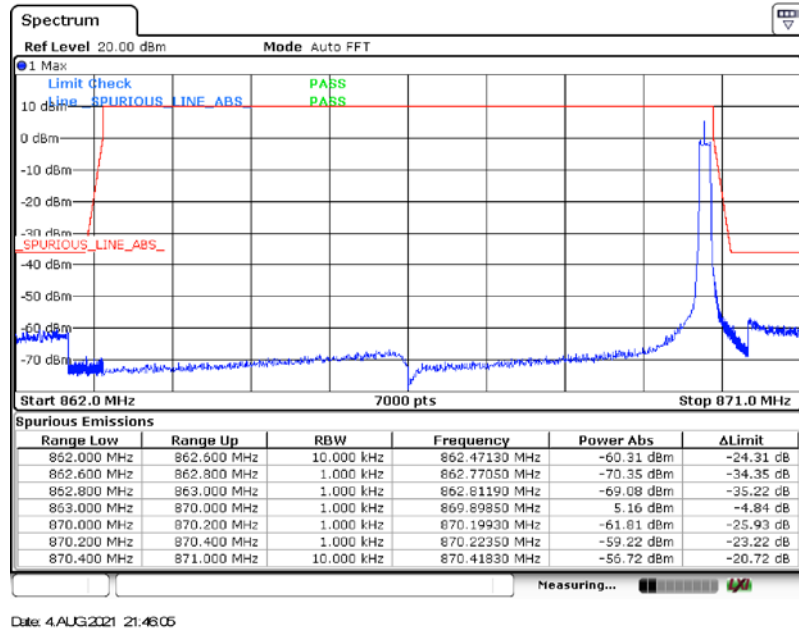


Out Of Band Domain for Operational Frequency Band

CH-L



CH-H



Appendix D: Transient power

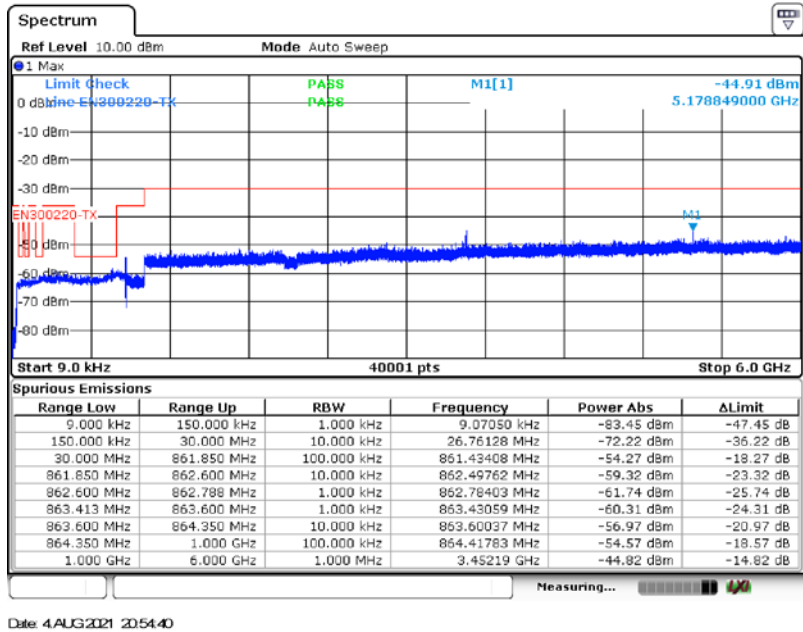
Test Channel	Offset from centre frequency	Reading (dBm)	Correct Factor (dB)	Transient power (dBm)	Limit (dBm)	Result
CH-L	-0.5 x OCW -1 200 kHz	-33.35	-24.8	-58.15	-27.00	Pass
	-0.5 x OCW - 400 kHz	-37.31	-20.0	-57.31	-27.00	Pass
	-OCW	-23.05	-13.0	-36.05	0.00	Pass
	-0.5 x OCW - 3 kHz	-13.56	0.0	-13.56	0.00	Pass
	0.5 x OCW + 3 kHz	-14.32	0.0	-14.32	0.00	Pass
	+OCW	-22.26	-13.0	-35.26	0.00	Pass
	0.5 x OCW + 400 kHz	-38.04	-20.0	-58.04	-27.00	Pass
	0.5 x OCW + 1 200 kHz	-33.31	-24.8	-58.11	-27.00	Pass
CH-M	-0.5 x OCW -1 200 kHz	-33.42	-24.8	-58.22	-27.00	Pass
	-0.5 x OCW - 400 kHz	-38.33	-20.0	-58.33	-27.00	Pass
	-OCW	-20.24	-13.0	-33.24	0.00	Pass
	-0.5 x OCW - 3 kHz	-13.06	0.0	-13.06	0.00	Pass
	0.5 x OCW + 3 kHz	-14.37	0.0	-14.37	0.00	Pass
	+OCW	-23.15	-13.0	-36.15	0.00	Pass
	0.5 x OCW + 400 kHz	-37.46	-20.0	-57.46	-27.00	Pass
	0.5 x OCW + 1 200 kHz	-33.57	-24.8	-58.37	-27.00	Pass
CH-H	-0.5 x OCW -1 200 kHz	-33.60	-24.8	-58.40	-27.00	Pass
	-0.5 x OCW - 400 kHz	-37.60	-20.0	-57.60	-27.00	Pass
	-OCW	-23.41	-13.0	-36.41	0.00	Pass
	-0.5 x OCW - 3 kHz	-12.62	0.0	-12.62	0.00	Pass
	0.5 x OCW + 3 kHz	-14.05	0.0	-14.05	0.00	Pass
	+OCW	23.05	-13.0	10.05	0.00	Pass
	0.5 x OCW + 400 kHz	-38.55	-20.0	-58.55	-27.00	Pass
	0.5 x OCW + 1 200 kHz	-33.13	-24.8	-57.93	-27.00	Pass

Note: Correct Factor= $10 \times \log(\text{RBWref}/\text{RBWmeas})$

Appendix E: Transmitter Unwanted Emissions in the Spurious Domain

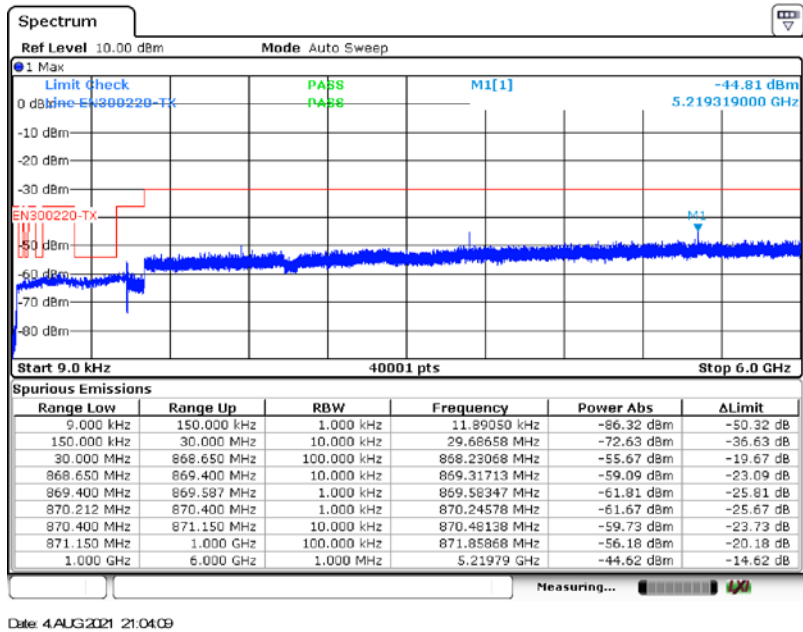
Test channel:

CH-L



Test channel:

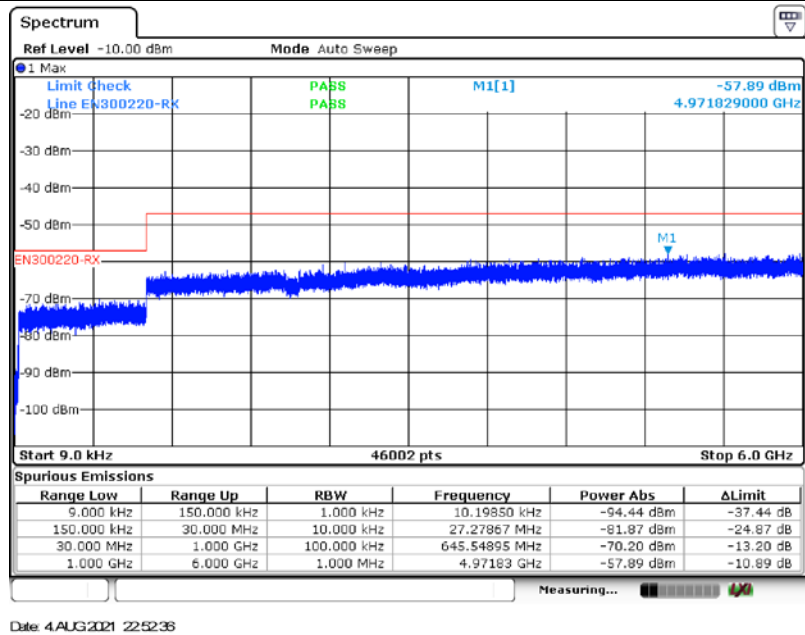
CH-H



Appendix F: Receiver Spurious Emissions

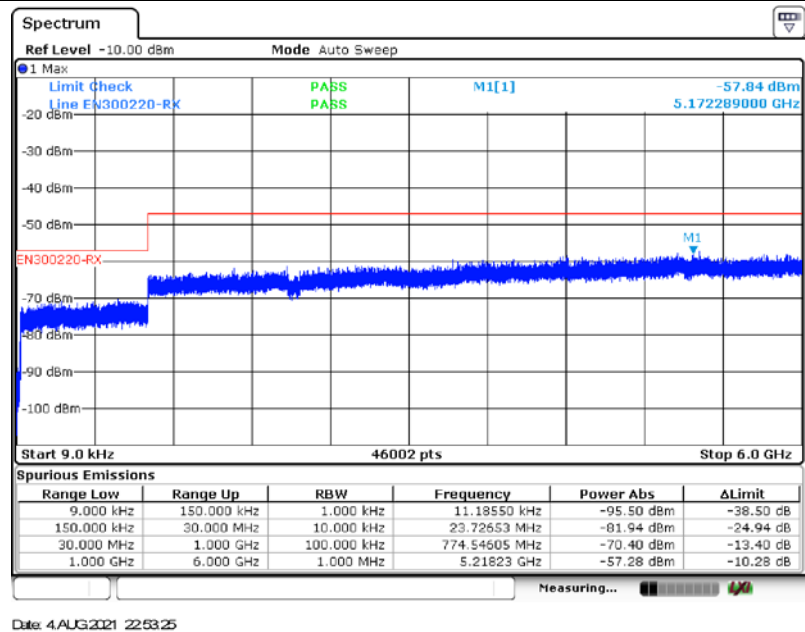
Test channel:

CH-L



Test channel:

CH-H



Appendix G: Blocking

Test channel	Wanted signal level / SGA ^{#1} (dBm)	Interference signal Offset centre frequency / SGB ^{#2} (MHz)	Interference signal Level / SGB (dBm)	Limit (dBm)	Result
CH-L	-93.03	-43.16	-35	≥-44	PASS
		-10	-33	≥-44	
		-2	-39	≥69	
		-2	-41	≥-69	
		-10	-33	≥-44	
		-43.16	-35	≥-44	
CH-M	-93.03	-43.42	-37	≥-44	PASS
		-10	-31	≥-44	
		-2	-40	≥69	
		-2	-38	≥-69	
		-10	-31	≥-44	
		-43.42	-36	≥-44	
CH-H	-93.03	-43.50	-35	≥-44	PASS
		-10	-30	≥-44	
		-2	-42	≥69	
		-2	-37	≥-69	
		-10	-33	≥-44	
		-43.50	-31	≥-44	

Note:

- #1: Signal generator A shall be set to the minimum level which gives the wanted performance criterion of EUT, the output level of generator A shall then be increased by 3 dB
- #2:

Receiver category	Interference signal frequency	Limit (dBm)
1	±2 MHz from Centre Frequency	≥ -20 dBm
	±10 MHz from Centre Frequency	
	±5 % of Centre Frequency or 15 MHz, whichever is the greater	
1.5	±2 MHz from OC edge f_{high} and f_{low}	≥ -43 dBm
	±10 MHz from OC edge f_{high} and f_{low}	≥ -33 dBm
	±5 % of Centre Frequency or 15 MHz, whichever is the greater	
2	±2 MHz from OC edge f_{high} and f_{low}	≥ -69 dBm
	±10 MHz from OC edge f_{high} and f_{low}	≥ -44 dBm
	±5 % of Centre Frequency or 15 MHz, whichever is the greater	
3	±2 MHz from OC edge f_{high} and f_{low}	≥ -80 dBm
	±10 MHz from OC edge f_{high} and f_{low}	≥ -60 dBm
	±5 % of Centre Frequency or 15 MHz, whichever is the greater	

-----End of Report-----